

Overview in General

This file contains a general overview of the data in the graph including node labels and relationships types.

References

- [jqassistant](#)
- [Neo4j Python Driver](#)

Node Labels

Table 1a - Highest node count by label combination

Lists the 30 label combinations with the highest number of nodes. The labels with the lowest node count are listed in table 1b. The total list would sum up to the total number of labels (100%).

The whole table can be found in the CSV report `Node_label_combination_count`.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Git, Update, Change]	57101	52.134177
1	[Git, Change, Create]	16651	15.202644
2	[Git, Commit]	11121	10.153661
3	[Git, Change, Delete]	8652	7.899422
4	[File, Git]	5754	5.253499
5	[Git, Change, Rename]	3288	3.002000
6	[Git, Tag]	1664	1.519260
7	[Author, Git, Person]	1257	1.147662
8	[Json, Key]	668	0.609895
9	[Json, Value, Scalar]	603	0.550549
10	[Committer, Git, Person]	370	0.337816
11	[NPM, Dependency]	338	0.308600
12	[Type, TS, Primitive]	291	0.265688
13	[Type, TS, Declared]	277	0.252906
14	[TS, ExternalDeclaration]	215	0.196299
15	[Type, TS, Literal]	136	0.124170
16	[Json, Value, Object]	133	0.121431
17	[Type, TS, Union]	119	0.108649
18	[Type, TS, ObjectMember]	102	0.093128
19	[NPM, Script]	91	0.083085
20	[Git, Change, Copy]	83	0.075780
21	[TS, Property]	65	0.059346
22	[TS, Function]	47	0.042912
23	[Type, TS, FunctionParameter]	40	0.036521
24	[Type, Object, TS]	39	0.035608
25	[Git, Branch]	38	0.034695
26	[File, Directory]	34	0.031043
27	[Type, TS, Function]	34	0.031043
28	[TS, Parameter]	33	0.030130
29	[Package, File, Json, NPM]	29	0.026477

Chart 1a - Highest node count by label combination

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in Chart 1b.

<Figure size 640x480 with 0 Axes>

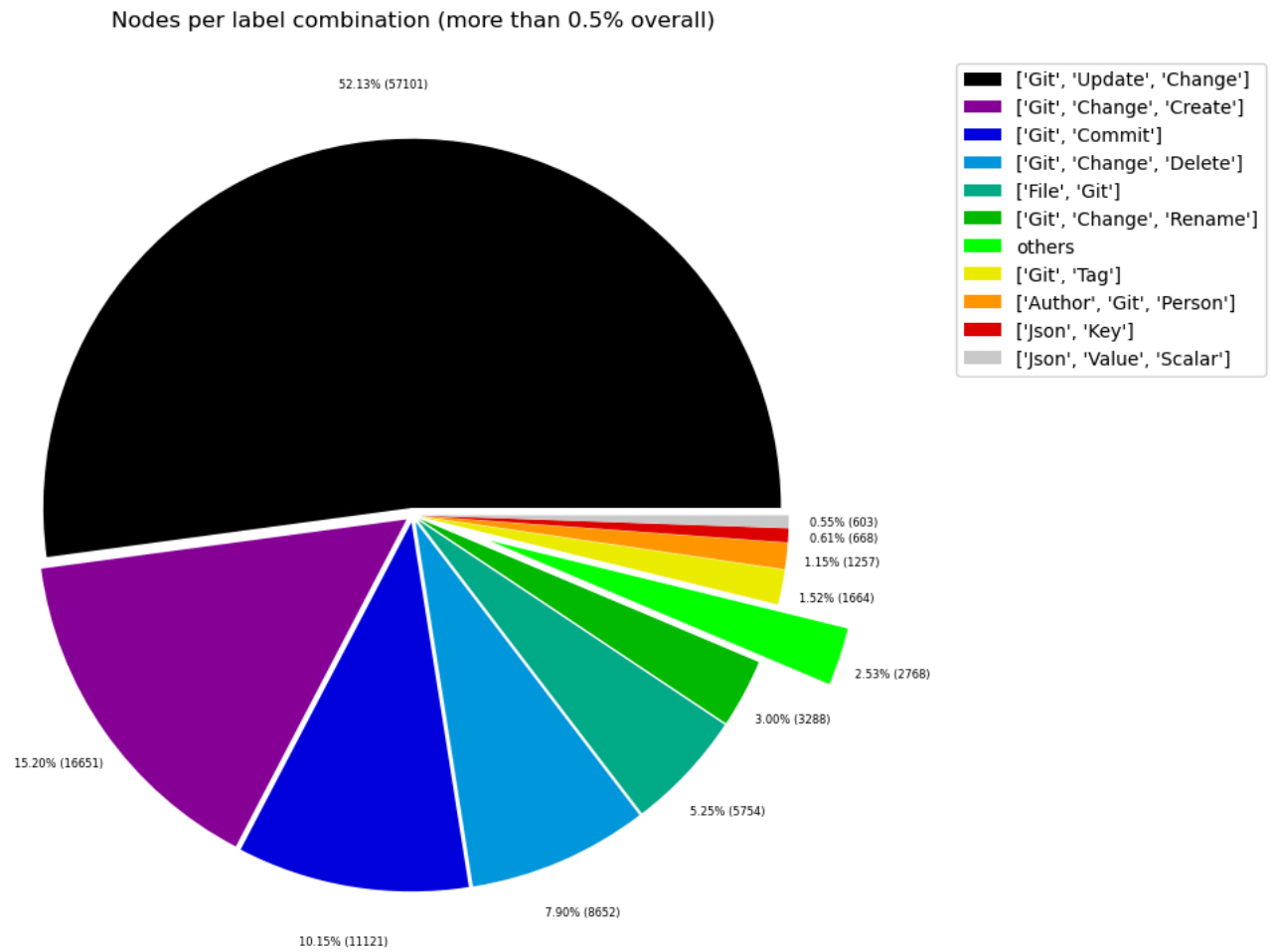


Table 1b - Lowest node count by label combination

Lists the 30 label combinations with the lowest number of nodes until they reach 0.5% of the total node count, which are shown above.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Analyze, Task, JQAssistant]	1	0.000913
1	[File, TS, Scan]	1	0.000913
2	[TS, Method]	1	0.000913
3	[Repository, File, Git]	1	0.000913
4	[TS, Constructor]	1	0.000913
5	[Value, TS, ObjectMember]	1	0.000913
6	[TS, Class]	1	0.000913
7	[TS, Enum]	2	0.001826
8	[Value, Object, TS]	3	0.002739
9	[Type, TS, Tuple]	3	0.002739
10	[Value, TS, Function]	4	0.003652
11	[TS, TypeParameter]	4	0.003652
12	[Value, TS, Complex]	5	0.004565
13	[NPM, Engine]	6	0.005478
14	[File, Local]	6	0.005478
15	[Project, TS]	6	0.005478
16	[Type, TS, TypeParameterReference]	6	0.005478
17	[Value, TS, Call]	6	0.005478
18	[Value, TS, Member]	6	0.005478
19	[File, TS, Local, Module]	6	0.005478
20	[TS, EnumMember]	8	0.007304
21	[Type, TS, NotIdentified]	11	0.010043
22	[TS, ExternalModule]	11	0.010043
23	[Json, Value, Array]	12	0.010956
24	[Value, TS, Declared]	13	0.011869
25	[TS, TypeAlias]	16	0.014608
26	[File, Directory, Local]	16	0.014608
27	[TS, Interface]	17	0.015521
28	[Type, TS, Intersection]	17	0.015521
29	[JQAssistant, Rule, Concept]	19	0.017347

Chart 1b - Lowest node count by label combination

Shows the lowest (less than 0.5% overall) node count label combinations. Therefore, this plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Nodes per label combination (less than 0.5% overall)

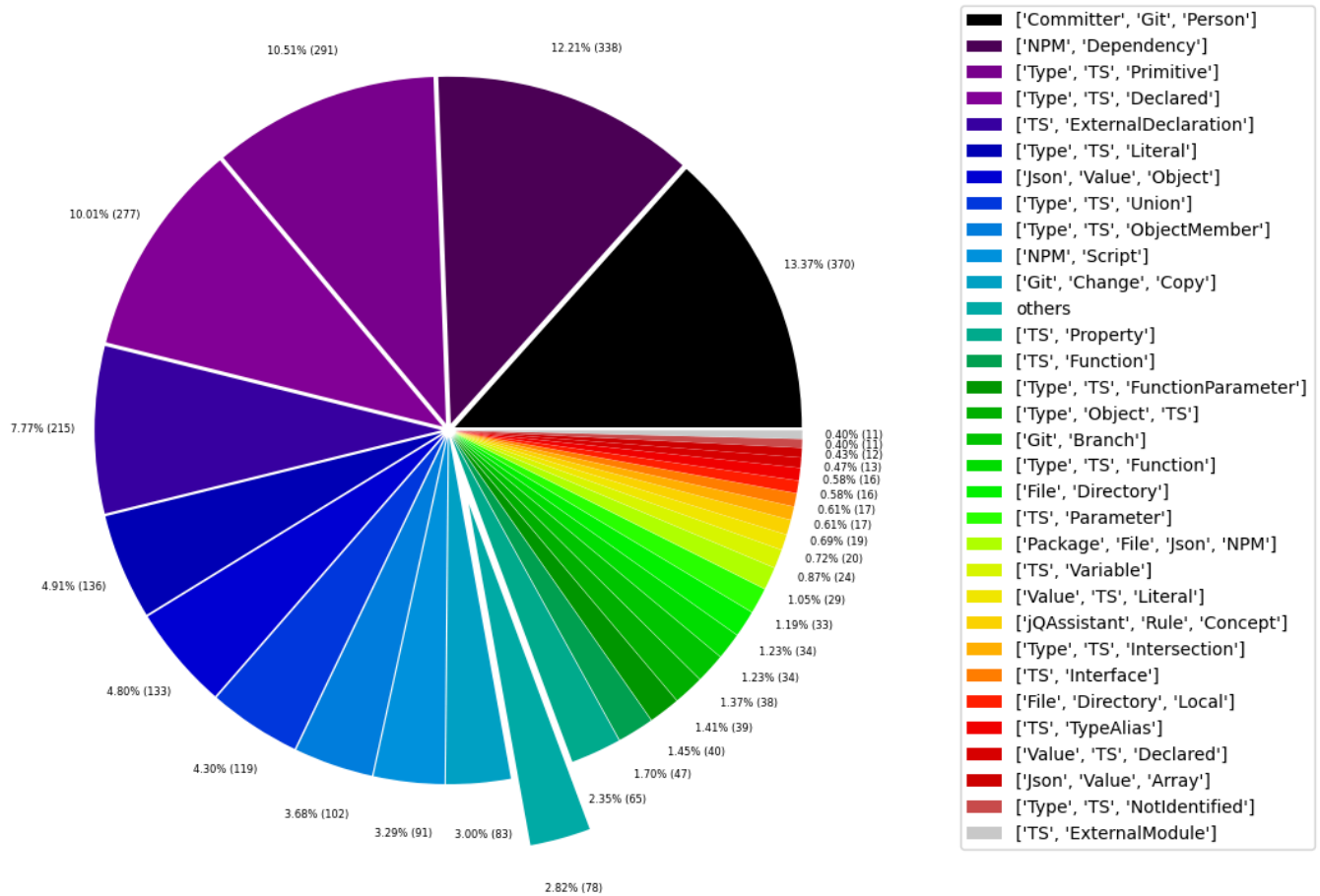


Table 1c - Highest node count by single label

Lists the 40 labels with the highest number of nodes. Doesn't sum up to the total number of nodes or 100% because one node can have multiple labels. Helps to identify commonly used labels.

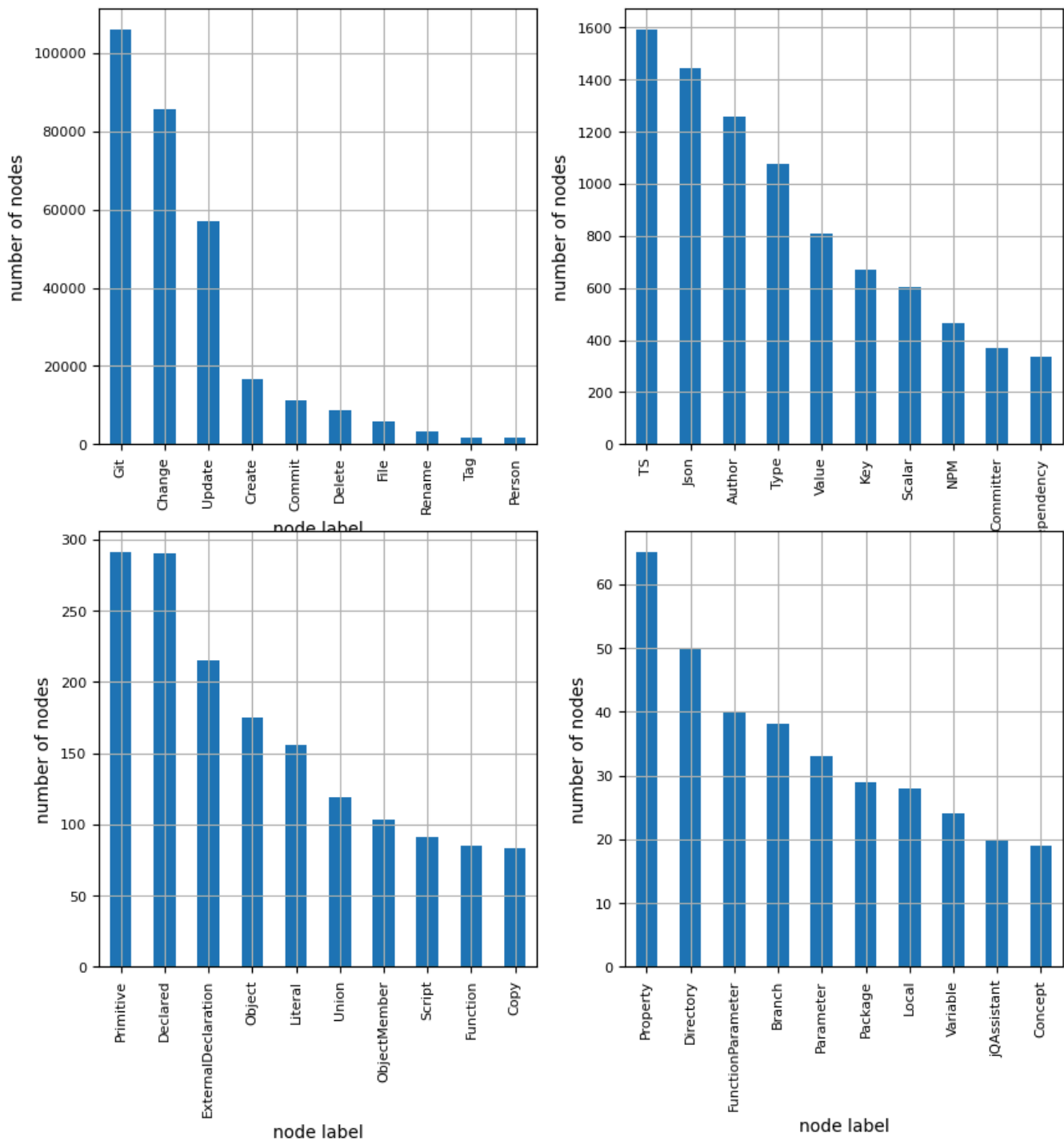
	nodeLabel	nodesWithThatLabel	nodesWithThatLabelPercent
0	Git	105980	96.761529
1	Change	85775	78.314023
2	Update	57101	52.134177
3	Create	16651	15.202644
4	Commit	11121	10.153661
5	Delete	8652	7.899422
6	File	5847	5.338410
7	Rename	3288	3.002000
8	Tag	1664	1.519260
9	Person	1627	1.485478
10	TS	1591	1.452610
11	Json	1445	1.319309
12	Author	1257	1.147662
13	Type	1075	0.981493
14	Value	806	0.735892
15	Key	668	0.609895
16	Scalar	603	0.550549
17	NPM	464	0.423640
18	Committer	370	0.337816
19	Dependency	338	0.308600
20	Primitive	291	0.265688
21	Declared	290	0.264775
22	ExternalDeclaration	215	0.196299
23	Object	175	0.159778
24	Literal	156	0.142431
25	Union	119	0.108649
26	ObjectMember	103	0.094041
27	Script	91	0.083085
28	Function	85	0.077606
29	Copy	83	0.075780
30	Property	65	0.059346
31	Directory	50	0.045651
32	FunctionParameter	40	0.036521
33	Branch	38	0.034695
34	Parameter	33	0.030130
35	Package	29	0.026477
36	Local	28	0.025564
37	Variable	24	0.021912
38	jQAssistant	20	0.018260
39	Concept	19	0.017347

Chart 1c - Highest node count by label

Shows the 40 labels with the highest number of nodes.

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Node count by label



Relationship Types

Table 2a - Highest relationship count by type

Lists the 30 relationship types with the highest number of occurrences. The whole table can be found in the CSV report `Relationship_type_count`.

Total number of relationships: 327044

	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	CONTAINS_CHANGE	85775	26.227358
1	MODIFIES	85775	26.227358
2	UPDATES	57101	17.459730
3	COMMITTED	22242	6.800920
4	CREATES	20022	6.122112
5	HAS_PARENT	12191	3.727633
6	DELETES	11940	3.650885
7	HAS_COMMIT	11121	3.400460
8	HAS_FILE	5754	1.759396
9	RENAMES	3288	1.005369
10	HAS_NEW_NAME	1763	0.539071
11	HAS_TAG	1664	0.508800
12	ON_COMMIT	1664	0.508800
13	HAS_AUTHOR	1257	0.384352
14	DEPENDS_ON	887	0.271217
15	HAS_KEY	668	0.204254
16	HAS_VALUE	668	0.204254
17	CONTAINS	594	0.181627
18	HAS_COMMITTER	370	0.113135
19	OF_TYPE	338	0.103350
20	EXPORTS	309	0.094483
21	REFERENCES	197	0.060237
22	DECLARES	186	0.056873
23	DECLARES_DEV_DEPENDENCY	169	0.051675
24	DECLARES_DEPENDENCY	161	0.049229
25	HAS_MEMBER	103	0.031494
26	HAS_TYPE_ARGUMENT	94	0.028742
27	DECLARES_SCRIPT	91	0.027825
28	COPIES	83	0.025379
29	RETURNS	82	0.025073

Chart 2a - Highest relationship count by type

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in the second chart.

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Relationship types (more than 0.5% overall)

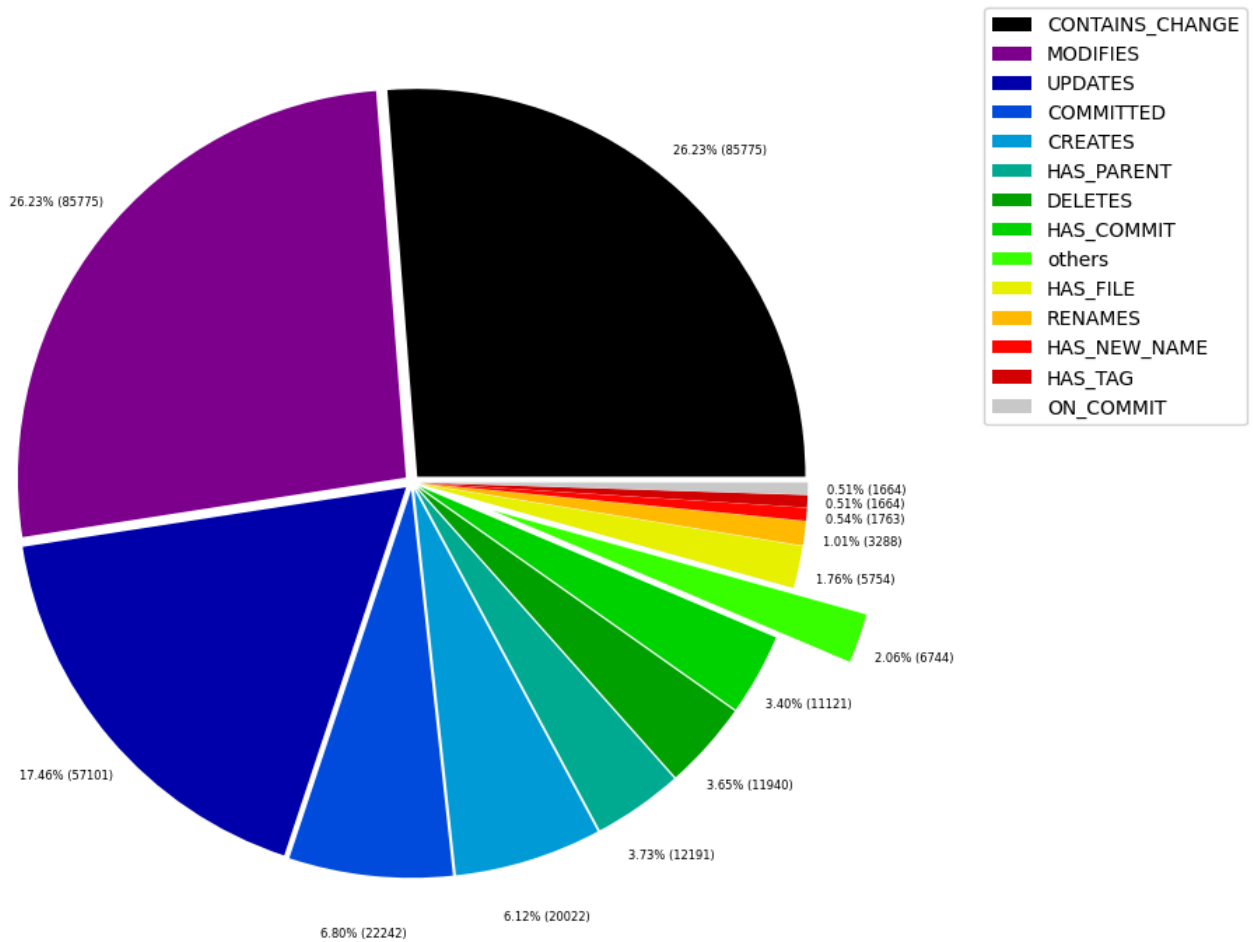


Table 2b - Lowest relationship count by type

Lists the 30 relationships type with the lowest number of occurrences up to 0.5% of the total node count. This is essentially breaking down the "others" slice from the chart above.

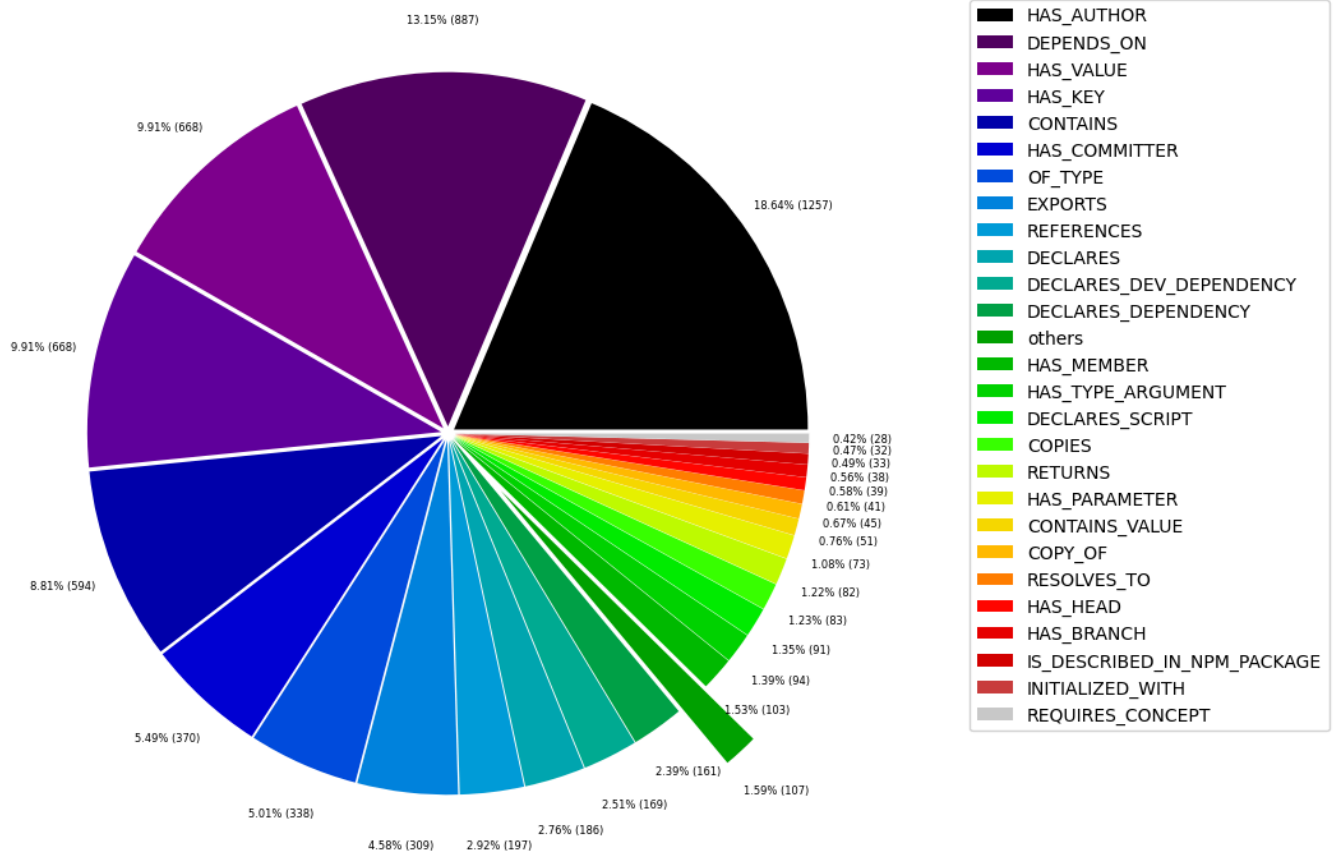
	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	CONSTRAINED_BY	4	0.001223
1	REFERENCED_PROJECTS	5	0.001529
2	MEMBER	6	0.001835
3	HAS_ROOT	6	0.001835
4	HAS_NPM_PACKAGE	6	0.001835
5	HAS_CONFIG	6	0.001835
6	HAS_ARGUMENT	6	0.001835
7	EXTENDS	6	0.001835
8	DECLARES_ENGINE	6	0.001835
9	CONTAINS_PROJECT	6	0.001835
10	CALLS	6	0.001835
11	PARENT	6	0.001835
12	DECLARES_PEER_DEPENDENCY	8	0.002446
13	USES	11	0.003363
14	INCLUDES_CONCEPT	19	0.005810
15	REQUIRES_CONCEPT	28	0.008562
16	INITIALIZED_WITH	32	0.009785
17	IS_DESCRIBED_IN_NPM_PACKAGE	33	0.010090
18	HAS_BRANCH	38	0.011619
19	HAS_HEAD	39	0.011925
20	RESOLVES_TO	41	0.012537
21	COPY_OF	45	0.013760
22	CONTAINS_VALUE	51	0.015594
23	HAS_PARAMETER	73	0.022321
24	RETURNS	82	0.025073
25	COPIES	83	0.025379
26	DECLARES_SCRIPT	91	0.027825
27	HAS_TYPE_ARGUMENT	94	0.028742
28	HAS_MEMBER	103	0.031494
29	DECLARES_DEPENDENCY	161	0.049229

Chart 2b - Lowest relationship count by type

Shows the lowest (less than 0.5% overall) relationship types. This plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Relationship types (less than 0.5% overall)



Node labels with their relationships

Table 3a - Highest relationship count by node labels and relationship type

Lists the 30 node labels and their relationship types with the highest number of occurrences.

	sourceLabels	relationType	targetLabels	numberOfRelationships	numberOfNodesWithSameLabelsAsSource	numberOf
0	[Git, Update, Change]	MODIFIES	[File, Git]	57101	57101	
1	[Git, Update, Change]	UPDATES	[File, Git]	57101	57101	
2	[Git, Commit]	CONTAINS_CHANGE	[Git, Update, Change]	57101	11121	
3	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Create]	16651	11121	
4	[Git, Change, Create]	MODIFIES	[File, Git]	16651	16651	
5	[Git, Change, Create]	CREATES	[File, Git]	16651	16651	
6	[Git, Commit]	HAS_PARENT	[Git, Commit]	12191	11121	
7	[Repository, File, Git]	HAS_COMMIT	[Git, Commit]	11121	1	
8	[Author, Git, Person]	COMMITTED	[Git, Commit]	11121	1257	
9	[Committer, Git, Person]	COMMITTED	[Git, Commit]	11121	370	
10	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Delete]	8652	11121	
11	[Git, Change, Delete]	DELETES	[File, Git]	8652	8652	
12	[Git, Change, Delete]	MODIFIES	[File, Git]	8652	8652	
13	[Repository, File, Git]	HAS_FILE	[File, Git]	5754	1	
14	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Rename]	3288	11121	
15	[Git, Change, Rename]	CREATES	[File, Git]	3288	3288	
16	[Git, Change, Rename]	DELETES	[File, Git]	3288	3288	
17	[Git, Change, Rename]	MODIFIES	[File, Git]	3288	3288	
18	[Git, Change, Rename]	RENAMES	[File, Git]	3288	3288	
19	[File, Git]	HAS_NEW_NAME	[File, Git]	1763	5754	
20	[Repository, File, Git]	HAS_TAG	[Git, Tag]	1664	1	
21	[Git, Tag]	ON_COMMIT	[Git, Commit]	1664	1664	
22	[Repository, File, Git]	HAS_AUTHOR	[Author, Git, Person]	1257	1	
23	[Json, Value, Object]	HAS_KEY	[Json, Key]	668	133	
24	[Json, Key]	HAS_VALUE	[Json, Value, Scalar]	552	668	
25	[Repository, File, Git]	HAS_COMMITTER	[Committer, Git, Person]	370	1	
26	[TS, Function]	DEPENDS_ON	[TS, ExternalDeclaration]	293	47	
27	[File, TS, Local, Module]	DEPENDS_ON	[TS, ExternalDeclaration]	236	6	
28	[TS, ExternalModule]	EXPORTS	[TS, ExternalDeclaration]	215	11	
29	[Package, File, Json, NPM]	DECLARES_DEV_DEPENDENCY	[NPM, Dependency]	169	29	

Graph Density

total_number_of_nodes (vertices): 109527

total_number_of_relationships (edges): 327044

-> total directed graph density: 2.726263106959576e-05

-> total directed graph density in percent: 0.002726263106959576